VIT

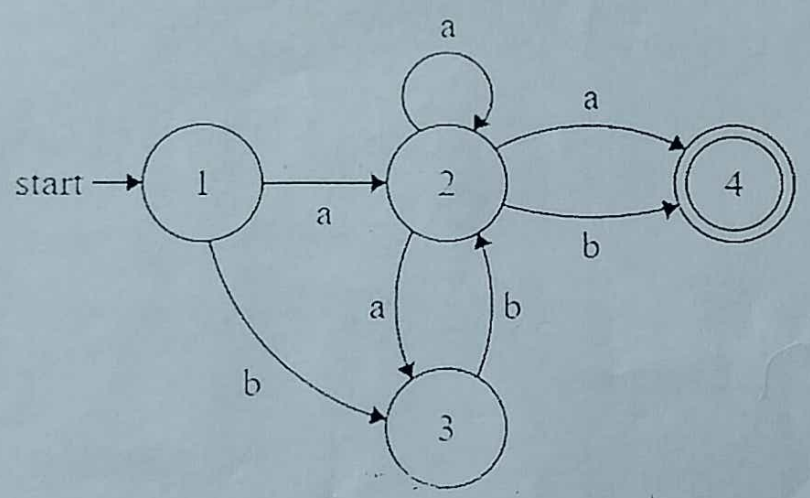
Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act 1956)

Continuous Assessment Test I – August 2024

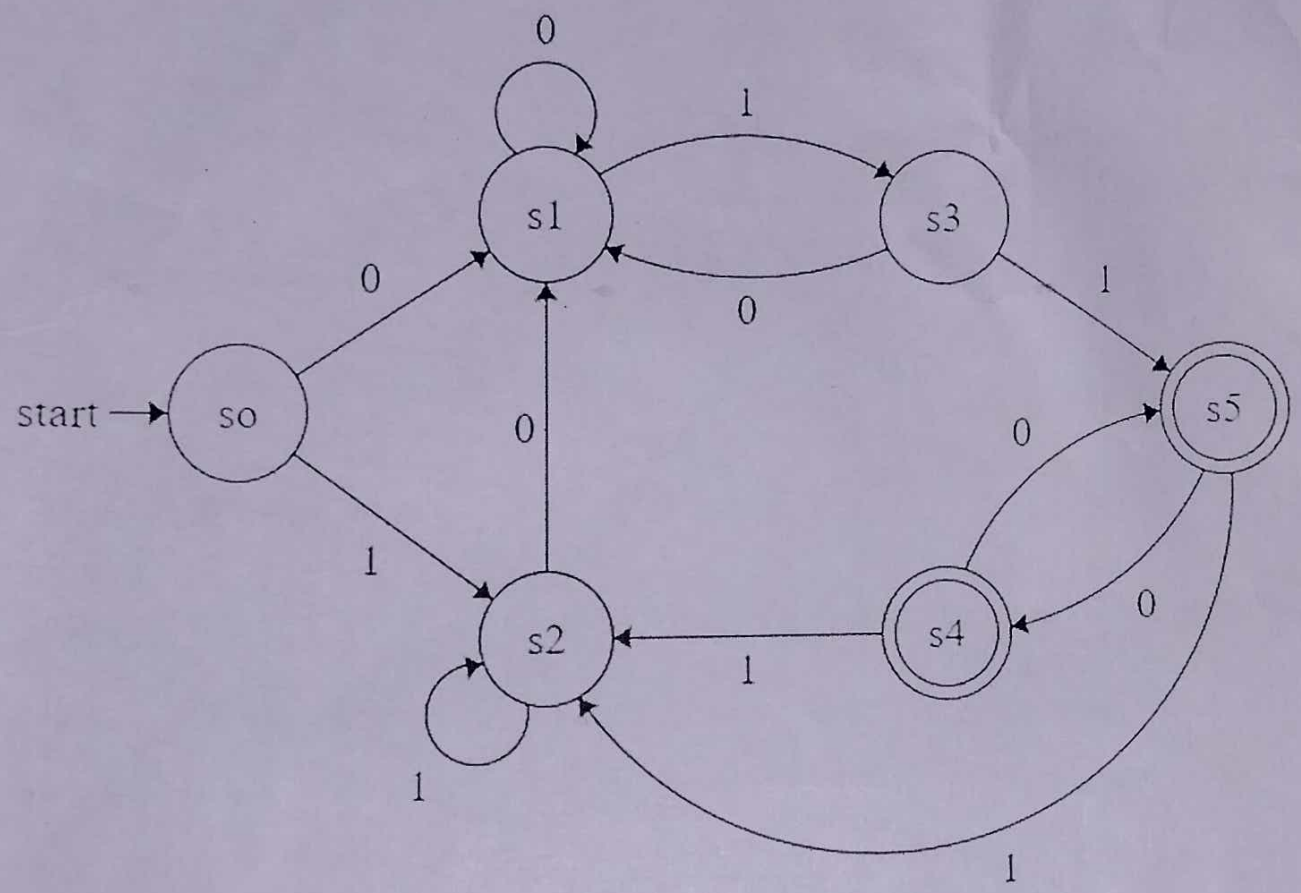
Programme	:	B.Tech Specialization AI, CPS, BRS, CSE core	Semester	:	Fall 2024-2025
Course Code	:	BCSE304L	Class Number	:	CH2024250100941 CH2024250102440, CH2024250100943, CH2024250100944
Course Title	:	Theory of Computation	Slot	:	C2+TC2
Faculty	:	Dr. SATHYARAJASEKARAN K, Dr. SHIVANI GUPTA, Dr. SUSEELA S, Dr. RENJITH			
Time	:	90 Minutes	Max. Marks	:	50

Answer ALL Questions

Q. No.	Question Text	Marks
1.	a) Prove by mathematical induction that 9^n-1 is divisible by 8 for all positive integers (7 Marks) b) Convert the following Finite State Automata (FSA) into an equivalent Regular expression (8 Marks)  	15
2.	a) Convert the Regular expression (a*|b*)* to Non-deterministic Finite State Automata (NFA) using Thompson's method (6 Marks) b) Construct an equivalent deterministic finite state automata DFA for the above constructed NFA in Question 2(a) (9 Marks)	15
3.	In a game of Mine craft, the outcome can either be a Win (W) or a Loss (L). The game either begins with one or more continuous series of wins and	10

concluded with two or more consecutive loss.

a) Find the Regular expression for the above game (3 Marks)
b) Construct a Finite State Automata (FSA) for the above expression (4 Marks)
c) Depict the transitions made by the constructed FSA for the input string WWLWLWLL (3 Marks)

4. Examine the DFA provided below and perform state minimization to reduce the number of states in the automata. Detail the steps involved in this reduction process. **10**

a) Steps for minimization (7 Marks)
b) Minimized DFA (3 Marks)